

Reasoning analysis — deepseek/deepseek-v4-pro (50 runs, real_deepseek_v4pro_50)

Each model rationale is split into sentences; sentences making the same point across runs are grouped. **Shared** = the point appears in at least 60% of the runs in that group; **varying** = it appears in some runs but not most. Counts are runs, not sentences. 'Evidence cited' lists the numbers and times the model quoted, with the number of runs quoting each.

decision: Profile / basal 00:00

keep — 28/50 runs

Varying reasoning

- (4/28) Keep and re-evaluate after ISF correction.
- (3/28) Overnight lows may be driven by residual IOB or SMB corrections rather than basal excess.
- (3/28) Change now risks masking the ISF effect.
- (2/28) Overnight hypoglycemia may be due to correction stacking rather than excessive basal; changing basal before addressing ISF and IOB limits could mask the true cause and risk worsening daytime control.
- (2/28) Overnight fasting glucose between 02:00-06:00 is stable with low TBR, and early-night hypos are driven by bolus-IOB, not excessive basal rate.
- (2/28) Overnight hypoglycemia is primarily driven by aggressive UAM/SMB and low effective ISF, not purely by basal rate.
- ... 4 more varying points

Points made by a single run only: 37.

Evidence cited: 06:00 (2), 02:00 (2), 4.8% (1), 90mg/dl (1), 10h (1), 105mg/dl (1), 18.5% (1), 01:00 (1), 03:00 (1), 00:00 (1), 0.6u/h (1), 0.5u/h (1).

change — 22/50 runs

Varying reasoning

- (8/22) Overnight lows and zero temp basals indicate basal excess.
- (6/22) Overnight hypoglycemia with TBR 18.5% at 01:00 indicates basal excess.
- (3/22) A 0.1 U/h reduction is a conservative first step.
- (2/22) Residual IOB from prior corrections is a confounder, so careful monitoring is required.
- (2/22) Reducing by 0.1 U/h is a safe first step.

Points made by a single run only: 31.

Phrases used by at least half the runs: "overnight hypoglycemia" (12).

Evidence cited: 18.5% (12), 0.5u/h (10), 0.1u/h (7), 01:00 (7), 18% (3), 0u/h (2), 90mg/dl (1), 90.6mg/dl (1), 2am (1), 0.6u/h (1).

decision: Profile / basal 05:00

keep — 49/50 runs

Varying reasoning

- (5/49) No clear hypoglycemia pattern in the 05:00-06:00 window; keep and re-evaluate after overnight basal is adjusted.

- (5/49) Change is not supported.
- (3/49) Keep until ISF fixed.
- (3/49) Early-morning glucose is stable with low TBR; no evidence for change.
- (2/49) Hypoglycemia risk decreases after 03:00, average glucose rises to ~106 mg/dL.
- (2/49) Keeping unchanged while monitoring effect of earlier reduction.
- ... 3 more varying points

Points made by a single run only: 52.

Evidence cited: 05:00 (9), 06:00 (3), 4.8% (2), 106mg/dl (2), 03:00 (2), 105mg/dl (2), 5% (2), 04:00 (2), 0.7u/h (1), 104.2mg/dl (1), 0.65u/h (1), 00:00 (1).

change — 1/50 runs

Points made by a single run only: 2.

Phrases used by at least half the runs: “glucose averages remain” (1), “overnight hypoglycemia risk” (1), “hypoglycemia risk period” (1), “segment maintains smooth” (1), “period glucose averages” (1), “lowering excess insulin” (1), “high reducing” (1), “transition while lowering” (1), “low and tbr” (1), “tbr is high” (1), “00 segment maintains” (1), “averages remain low” (1).

Evidence cited: 00:00 (1).

decision: Profile / basal 06:00

keep — 50/50 runs

Varying reasoning

- (6/50) Hourly glucose at 06:00 averages 99.5 mg/dL with TBR 1.3% and no evidence of hypoglycemia.
- (4/50) Morning rise is well controlled with minimal hypoglycemia.
- (3/50) Morning rise may be due to dawn phenomenon or unannounced meals; without meal data, increasing basal could be unsafe.
- (3/50) Insufficient evidence to change; morning rise may be meal-related, not basal-deficient.
- (3/50) Daytime basal appears adequate; no isolated data to support change.
- (3/50) Morning glucose is well controlled (TIR>91%) with this rate.
- ... 5 more varying points

Points made by a single run only: 41.

Evidence cited: 06:00 (7), 1.3% (5), 99.5mg/dl (3), 07:00 (3), 08:00 (2), 91% (2), 0.8u/h (1), 2.5% (1), 98.7% (1), 110.5mg/dl (1), 100mg/dl (1), 10:00 (1).

decision: Profile / basal 10:00

keep — 50/50 runs

Varying reasoning

- (3/50) Keep pending meal logging.
- (3/50) Daytime TIR remains acceptable; afternoon hyperglycemia likely meal-driven, not basal.
- (2/50) Daytime basal appears adequate; observed highs are likely meal-related (no carb data).
- (2/50) Daytime patterns are confounded by unannounced meals; insufficient evidence to adjust.

- (2/50) Daytime basal adequacy cannot be isolated from unknown meal and activity effects.
- (2/50) No clear pattern requiring adjustment; mixed with meal and correction effects.
- ... 5 more varying points

Points made by a single run only: 40.

Evidence cited: 0.7u/h (1), 148.0mg/dl (1), 0.8u/h (1), 20.2% (1), 06:00 (1), 13:00 (1), 16:00 (1).

decision: Profile / basal 16:00

keep — 50/50 runs

Varying reasoning

- (3/50) This is the lowest daytime basal; afternoon hyperglycemia may be meal-driven rather than basal deficiency.
- (3/50) Lowest basal rate; coincides with typical afternoon activity.
- (3/50) Keep until meal-data available.
- (2/50) The high TAR at this time could be due to missing meal boluses or the basal drop, but without carb logs a change is not justified.
- (2/50) Although late afternoon glucose remains high, increasing basal without meal context could worsen evening lows.
- (2/50) Late-afternoon hyperglycemia exists but may be meal-related; no evidence to change.
- ... 2 more varying points

Points made by a single run only: 56.

Evidence cited: 0.5u/h (5), 18:00 (3), 19:00 (2), 15.3% (2), 32.1% (2), 16:00 (2), 148.9mg/dl (1), 32% (1), 163mg/dl (1).

decision: Profile / basal 18:00

keep — 50/50 runs

Varying reasoning

- (2/50) Similar to the 16:00 segment, the interaction with possible evening meals and the subsequent 0.7 U/h segment at 19:00 is complex.
- (2/50) Keep until meal data clarifies need.
- (2/50) Evening lows attributed to IOB, not basal rate.
- (2/50) Evening basal appears reasonable; no strong hypo pattern.
- (2/50) No strong evidence for change; evening lows likely IOB-related.
- (2/50) No independent evidence that basal here causes lows; corrections are the primary driver.
- ... 1 more varying points

Points made by a single run only: 49.

Evidence cited: 18:00 (4), 7.7% (3), 16:00 (3), 0.7u/h (2), 19:00 (2), 0.5u/h (1), 124.2mg/dl (1), 0.6u/h (1).

decision: Profile / basal 19:00

keep — 48/50 runs

Varying reasoning

- (3/48) Evening basal appears appropriate.
- (3/48) Evening basal appears adequate; no change supported.
- (3/48) Evening glucose stable; no change indicated.
- (2/48) Similar to 18:00, no isolated fasting window exists to test this segment.
- (2/48) Evening pattern is acceptable; too early to change without meal data.
- (2/48) Evening profile segment; no independent evidence of basal inappropriateness.

Points made by a single run only: 44.

Evidence cited: 19:00 (4), 18:00 (3), 15.3% (2), 4.8% (1), 0.7u/h (1), 120.0mg/dl (1), 20:00 (1).

change — 2/50 runs

Points made by a single run only: 4.

Phrases used by at least half the runs: “overnight basal” (2), “tbr 15.3” (2), “0.6 u/h” (1), “post 18 00” (1), “overnight basal reduction” (1), “drop this change” (1), “basal segment reduction” (1), “lessen the post” (1), “reduction is stable” (1), “00 tbr 15.3” (1), “u/h is expected” (1), “evening hypoglycemia 19” (1).

Evidence cited: 15.3% (2), 19:00 (2), 18:00 (1), 0.6u/h (1), 0.1u/h (1), 105mg/dl (1).

decision: Profile / CR 00:00

keep — 50/50 runs

Varying reasoning

- (15/50) CR cannot be evaluated.
- (12/50) Keep until meal logging provides data.
- (7/50) No carbohydrate entries exist to evaluate this ratio.
- (4/50) Cannot be validated without carb entries.
- (4/50) No meal data exist to evaluate any CR segment.
- (3/50) Cannot evaluate without meal data; keep until meals are logged.
- ... 6 more varying points

Points made by a single run only: 20.

Evidence cited: 13g/u (1).

decision: Profile / CR 04:00

keep — 50/50 runs

Varying reasoning

- (5/50) No carb data; keep.
- (4/50) No meal data to assess.

- (4/50) No carb entries.
- (3/50) Keep for now; re-evaluate after meals are logged.
- (3/50) No meal data available.
- (2/50) No meal entries available; carb ratio cannot be evaluated.
- ... 4 more varying points

Points made by a single run only: 15.

Evidence cited: 11.1g/u (1).

decision: Profile / CR 08:00

keep — 50/50 runs

Varying reasoning

- (5/50) No carb data; keep.
- (4/50) Cannot evaluate without carb entries.
- (4/50) No meal data to assess.
- (3/50) No carbohydrate entries.
- (3/50) No meal data; keep until validated.
- (2/50) CR cannot be evaluated.
- ... 2 more varying points

Points made by a single run only: 12.

Evidence cited: 13.2g/u (1).

decision: Profile / CR 16:00

keep — 50/50 runs

Varying reasoning

- (5/50) Cannot evaluate without carb entries.
- (4/50) No carb data; keep.
- (3/50) No meal data available to assess.
- (2/50) CR cannot be evaluated.
- (2/50) Cannot assess; keep until meal data present.
- (2/50) No meal data; keep.
- ... 1 more varying points

Points made by a single run only: 18.

Evidence cited: 10.5g/u (1).

decision: Profile / CR 20:00

keep — 50/50 runs

Varying reasoning

- (5/50) No carb data; keep.
- (4/50) Cannot evaluate without carb entries.
- (3/50) No meal data available to assess.
- (2/50) Evening meal pattern unconfirmed; no change without meal logs.
- (2/50) No meal entries available; carb ratio cannot be evaluated.
- (2/50) No meal data; keep.
- ... 1 more varying points

Points made by a single run only: 15.

Evidence cited: 127.6mg/dl (1), 12.6g/u (1).

decision: Profile / ISF 00:00

keep — 30/50 runs

Varying reasoning

- (4/30) No change needed.
- (2/30) Dynamic ISF continuously adjusts sensitivity; altering the static ISF now would be premature without meal data and after basal correction.
- (2/30) However, changing ISF now would complicate basal adjustments; keep and re-evaluate after basal correction and carb logging.
- (2/30) Dynamic ISF compensates; static ISF cannot be reliably adjusted without meal data and a full sensitivity test.
- (2/30) Dynamic ISF overrides static ISF; observed sensitivity range handled adequately.
- (2/30) Dynamic ISF (70%) aggressively adapts; profile ISF not the primary cause of lows/highs.
- ... 2 more varying points

Points made by a single run only: 38.

Phrases used by at least half the runs: “dynamic isf” (25).

Evidence cited: 70% (2), 56mg/dl (1), 124mg/dl (1), 40mg/dl (1), 0.2u (1), 178mg/dl (1).

change — 20/50 runs

Varying reasoning

- (4/20) Multiple correction bolus windows show drops 5-10x larger than profile ISF predicts, indicating actual sensitivity is much higher.
- (2/20) Observed correction overshoots and low variable_sens values indicate effective ISF is too low; increasing ISF reduces correction strength and lowers hypoglycemia risk.
- (2/20) SMB and correction doses produce excessive glucose drops.
- (2/20) Observed correction bolus drops greatly exceed predictions of 56 mg/dL/U.
- (2/20) 30 mg/dL/U is a safer starting value that reduces IOB stacking and nocturnal hypoglycemia.
- (2/20) This is the single most impactful safe profile change supported by the data.

Points made by a single run only: 41.

Evidence cited: 56mg/dl (5), 70mg/dl (3), 110mg/dl (2), 0.2u (2), 178mg/dl (2), 27.8mg/dl (2), 100mg/dl (2), 30mg/dl (1), 53mg/dl (1), 150mg/dl (1), 36mg/dl (1), 38mg/dl (1).

decision: Profile / target 00:00

keep — 49/50 runs

Varying reasoning

- (9/49) Target of 90 mg/dL is reasonable.
- (5/49) Changing the target would not address the root cause.
- (4/49) Target is appropriate; hypoglycemia stems from basal excess, not target.
- (3/49) Reducing basal is the first-line measure; target increase may be considered later if lows persist.
- (3/49) Target is appropriate; hypoglycemia is from over-delivery, not target level.
- (2/49) While an overnight target of 90 may be aggressive given frequent lows, changing ISF will indirectly reduce hypoglycemia.
- ... 4 more varying points

Points made by a single run only: 42.

Evidence cited: 90mg/dl (13), 100mg/dl (2), 105mg/dl (1).

change — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “overnight target contributes” (1), “control evidence hourly_glucose_local” (1), “targeting good control” (1), “early morning hours” (1), “still targeting good” (1), “90 mg/dl” (1), “18.5 in early” (1), “hourly_glucose_local tbr” (1), “provides safety buffer” (1), “mg/dl provides safety” (1), “hypoglycemia events raising” (1), “100 mg/dl provides” (1).

Evidence cited: 90mg/dl (1), 100mg/dl (1), 18.5% (1).

decision: Profile / target 08:00

keep — 50/50 runs

Varying reasoning

- (8/50) Daytime target of 100 mg/dL is appropriate.
- (5/50) No change needed.
- (5/50) Daytime target reasonable.
- (5/50) Daytime target is appropriate.
- (3/50) Daytime target appears appropriate; no data to contradict.
- (2/50) No evidence that this is causing hypo- or hyperglycemia beyond what would be explained by meal or basal issues.
- ... 2 more varying points

Points made by a single run only: 24.

Phrases used by at least half the runs: “daytime target” (43).

Evidence cited: 100mg/dl (11), 148mg/dl (1).

decision: Profile / target 23:00

keep — 49/50 runs

Varying reasoning

- (8/49) Same as 00:00 target.
- (8/49) Evening target is appropriate.
- (6/49) Consistent with overnight target.
- (2/49) Evening target back to 90 mg/dL; consistent and safe after DIA fix.
- (2/49) Returns to 90 mg/dL at 23:00, aligning with early-night hours.
- (2/49) Target consistent with overnight needs.

Points made by a single run only: 24.

Evidence cited: 00:00 (8), 90mg/dl (5), 23:00 (3), 10.1% (1), 105.7mg/dl (1).

change — 1/50 runs

Points made by a single run only: 1.

Phrases used by at least half the runs: “sleep transition” (1), “mg/dl further reduces” (1), “overnight target” (1), “reduces hypoglycemia risk” (1), “100 mg/dl” (1), “late night target” (1), “target of 100” (1), “aligning the late” (1).

Evidence cited: 100mg/dl (1).

decision: Profile / DIA

keep — 42/50 runs

Varying reasoning

- (4/42) Re-evaluate after ISF fix and with meal logging.
- (3/42) Insulin action duration in correction windows is consistent with a 10-h DIA; no stacking or early washout observed.
- (3/42) No evidence to suggest a DIA mismatch; cannot test without meal boluses.
- (2/42) Observed correction action times are consistent with 10-hour DIA; no evidence of insulin stacking or premature tail-off.
- (2/42) Long DIA may overestimate IOB but there is no evidence of harm; changing it would affect IOB calculations and could destabilise the system.
- (2/42) Standard DIA for rapid-acting insulin; no evidence of insulin stacking or tail mismatch.
- ... 4 more varying points

Points made by a single run only: 45.

Evidence cited: 10h (13), 00:00 (1), 02:00 (1), 9h (1), 4h (1), 6h (1).

change — 8/50 runs

Varying reasoning

- (3/8) 5 h matches typical rapid-acting insulin duration.

Points made by a single run only: 17.

Phrases used by at least half the runs: “rapid acting insulin” (5), “insulin action” (4).

Evidence cited: 10h (4), 6h (4), 5h (2), 8h (1), 7h (1).

decision: AAPS / SMB and UAM

keep — 49/50 runs

Varying reasoning

- (8/49) SMB and UAM are essential for unannounced meal coverage.
- (4/49) Both are essential for closed-loop function; no adverse effects detected.
- (4/49) Both features are essential for the current automation strategy.
- (3/49) Both are required for closed-loop operation.
- (2/49) Disabling SMB or UAM could remove the primary insulin delivery mechanism for unannounced meals.
- (2/49) No configuration issue detected; hypoglycemia stems from ISF miscalibration.
- ... 5 more varying points

Points made by a single run only: 56.

change — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “enabled for correction” (1), “meals and feeds” (1), “misinterprets unlogged meals” (1), “uam is safety” (1), “safety requirement” (1), “isf disabling uam” (1), “aggressive smb” (1), “correction boluses” (1), “smb under dynamic” (1), “meal logging” (1), “dynamic isf disabling” (1), “requirement while meal” (1).

decision: AAPS / SMB activation conditions

keep — 48/50 runs

Varying reasoning

- (4/48) SMB during temp targets is acceptable;
- (4/48) Current activation conditions are safe and conservative for a system with no carb entries.
- (3/48) No change needed.
- (3/48) Activation rules are appropriate: SMB always on ensures prompt response;
- (3/48) May reconsider smb_with_cob_enabled after meal logging begins, but no change now.
- (3/48) SMB with temp target enabled is appropriate.
- ... 6 more varying points

Points made by a single run only: 50.

change — 2/50 runs

Points made by a single run only: 8.

Phrases used by at least half the runs: “135 mg/dl” (1), “likely for exercise” (1), “temp target period” (1), “conditions are appropriate” (1), “uam driven meal” (1), “appropriate since” (1), “changed from true” (1), “user set six” (1), “requests reduced insulin” (1), “false is appropriate” (1), “true to false” (1), “targets of 135” (1).

Evidence cited: 66mg/dl (1), 135mg/dl (1).

decision: AAPS / SMB strength and frequency

keep — 45/50 runs

Varying reasoning

- (3/45) After ISF correction, these limits should be safe.
- (3/45) Values within typical ranges.
- (3/45) No change needed.
- (2/45) Delivery parameters are moderate; the root cause of excessive insulin is not the interval but the per-unit effect.
- (2/45) Standard values that do not independently cause hypoglycemia.
- (2/45) SMB doses are small, but the ISF issue makes them overly powerful.
- ... 3 more varying points

Points made by a single run only: 51.

Evidence cited: 0.27u (1), 0.8u/h (1), 0.4u (1), 0.3u (1).

change — 5/50 runs

Points made by a single run only: 10.

Phrases used by at least half the runs: “basal minutes” (3).

Evidence cited: 20min (2), 10min (2), 15min (2).

decision: AAPS / max basal and max IOB limits

keep — 28/50 runs

Varying reasoning

- (8/28) Limits are far above actual usage; no safety concern.
- (3/28) Limits are never reached; no change needed.
- (2/28) Highest observed temp basal: 3.6 U/h.
- (2/28) These are high relative to actual daily insulin use but no evidence of near-limit activation.
- (2/28) Generous limits; not constraining therapy or causing risk.
- (2/28) These are safety ceilings and are not restricting therapy.
- ... 2 more varying points

Points made by a single run only: 20.

Evidence cited: 3.6u/h (2), 25u (1), 5.935u (1), 12u/h (1), 6u (1), 1u/h (1), 5.9u (1).

change — 22/50 runs

Varying reasoning

- (5/22) Max IOB of 25 U is far beyond any safe need (total daily insulin ~20 U);
- (4/22) Max basal at 12 U/h is acceptable.
- (3/22) Lowering max basal to 4 U/h and max IOB to 5 U creates safe ceilings that prevent massive insulin accumulation.
- (3/22) Reducing max IOB to 10 U provides a meaningful safety cap while still permitting necessary corrections.
- (2/22) Observed max temp basal=3.6 U/h; a limit of 4 U/h allows all needed delivery while preventing extreme rates.
- (2/22) Max IOB of 25 U is dangerous given extreme sensitivity; reducing it to 2.0 U prevents catastrophic insulin stacking.
- ... 1 more varying points

Points made by a single run only: 39.

Phrases used by at least half the runs: “max iob” (15), “max basal” (12).

Evidence cited: 25u (15), 12u/h (9), 5u (7), 6u (6), 8u (5), 10u (4), 20u (4), 4u/h (4), 3u/h (2), 3.6u/h (2), 19.7u (2), 2u/h (2).

decision: AAPS / Autosens and Dynamic ISF

keep — 41/50 runs

Varying reasoning

- (4/41) Keep and re-evaluate after basal and max IOB changes.
- (3/41) Dynamic ISF effectively compensates for profile ISF; no adverse target adjustments.
- (2/41) 70% damping is a reasonable starting point.
- (2/41) Dynamic ISF is functioning as intended; any aggressive action results from underlying profile ISF rather than this setting.
- (2/41) Dynamic ISF is handling sensitivity changes appropriately; no need to alter the 70% factor or enable Autosens.
- (2/41) The root cause is the base ISF, not the adjustment factor.
- ... 4 more varying points

Points made by a single run only: 51.

Phrases used by at least half the runs: “dynamic isf” (40).

Evidence cited: 70% (8).

change — 9/50 runs

Varying reasoning

- (2/9) The 70% floor enables an effective ISF as low as ~39 mg/dL/U, causing unsafe over-correction and contributing to severe hypoglycaemia.
- (2/9) Dynamic ISF with 70% adjustment factor is causing dangerous over-amplification of insulin sensitivity.

Points made by a single run only: 20.

Phrases used by at least half the runs: “dynamic isf” (5), “effective isf” (5).

Evidence cited: 70% (8), 100% (4), 30% (2), 50% (2), 16.8mg/dl (1), 40% (1), 27.8mg/dl (1), 60% (1), 42mg/dl (1), 41mg/dl (1), 22.4mg/dl (1), 85% (1).

decision: AAPS / carbohydrate absorption model

keep — 46/50 runs

Varying reasoning

- (5/46) No carb entries to validate absorption parameters; keep defaults until carb logging is established.
- (3/46) Re-evaluate after meal logging.
- (3/46) Defaults are acceptable; no meal data to evaluate performance.
- (3/46) No carb data to evaluate; keep unchanged.
- (2/46) Review after meal logging begins to ensure they match observed absorption.
- (2/46) Absorption model may exaggerate UAM response, but without meal data, change is unsupported.
- ... 7 more varying points

Points made by a single run only: 36.

Evidence cited: 5min (1), 6h (1), 8mg/dl (1).

change — 3/50 runs

Points made by a single run only: 8.

Phrases used by at least half the runs: “mg/dl/ min” (3), “deliver excessive” (2), “unannounced meals” (2).

Evidence cited: 5min (3), 5mg/dl (2), 8mg/dl (2), 5h (1), 3mg/dl (1).

verify — 1/50 runs

Points made by a single run only: 1.

Phrases used by at least half the runs: “hours shortens iob” (1), “logging begins assess” (1), “horizon may prolong” (1), “assess whether reduction” (1), “begins assess whether” (1), “iob after meal” (1), “tails without compromising” (1), “compromising postprandial coverage” (1), “shortens iob tails” (1), “meal logging begins” (1), “prolong uam iob” (1), “reduction to hours” (1).

Evidence cited: 6h (1).

profile recommendation

basal — 28/50 runs

Shared reasoning

- (21/28) Reduce the basal rate from 00:00 to 05:00 from 0.6 U/h to 0.5 U/h.

Varying reasoning

- (7/28) Hourly local glucose shows average 90.6 mg/dL and TBR 18.5% at 01:00 despite the loop frequently running zero temp basals.
- (7/28) Keep all other basal segments unchanged.
- (4/28) A 0.1 U/h reduction (16.7%) is a conservative first step to mitigate hypoglycemia without risking hyperglycemia.
- (3/28) Keep all current profile settings (basal, CR, ISF, targets, DIA) unchanged.

- (3/28) A 0.1 U/h reduction is a conservative first step that directly addresses the dominant hypoglycemia risk.
- (2/28) Hourly glucose data shows peak hypoglycemia (18.5% TBR) at 01:00 local when that basal has been active for several hours and meal IOB is minimal.
- ... 3 more varying points

Points made by a single run only: 57.

Phrases used by at least half the runs: “0.5 u/h” (18), “basal rate” (17), “0.6 u/h” (16).

Evidence cited: 00:00 (21), 05:00 (19), 0.5u/h (18), 0.6u/h (16), 18.5% (12), 0.1u/h (8), 01:00 (7), 0u/h (4), 0.55u/h (3), 90mg/dl (2), 90.6mg/dl (2), 06:00 (1).

isf — 17/50 runs

Varying reasoning

- (6/17) Increase ISF from 56 mg/dL/U to 150 mg/dL/U.
- (2/17) Lower the ISF from 56 mg/dL/U to 30 mg/dL/U as an initial trial value, reflecting the much greater insulin sensitivity observed in correction responses and the dynamic ISF outputs.
- (2/17) Correction boluses of 0.05-0.2 U result in glucose drops of 26-178 mg/dL, far exceeding the 56 mg/dL/U prediction.
- (2/17) Observed correction bolus drops (e.g., -125 mg/dL on 0.2 U) are 5-10x the expected drop using current ISF, indicating actual insulin sensitivity is much higher.
- (2/17) Increase the Insulin Sensitivity Factor from 56 mg/dL/U to 70 mg/dL/U to reduce correction strength and mitigate hypoglycemia.
- (2/17) Multiple correction boluses produced drops > 2-3 times what the current ISF predicts.

Points made by a single run only: 52.

Phrases used by at least half the runs: “56 mg/dl/u” (16), “isf from 56” (15), “dynamic isf” (10), “insulin sensitivity” (10).

Evidence cited: 56mg/dl (16), 0.2u (6), 70mg/dl (4), 100mg/dl (3), 30mg/dl (2), 178mg/dl (2), 50mg/dl (2), 40mg/dl (2), 06:00 (1), 4.8% (1), 13.2g/u (1), 02:00 (1).

dia — 4/50 runs

Varying reasoning

- (2/4) Reduce DIA from 10 hours to 6 hours.

Points made by a single run only: 12.

Phrases used by at least half the runs: “dia from 10” (3), “10 hours” (2), “reduce dia” (2), “hour dia” (2), “insulin action” (2).

Evidence cited: 10h (4), 6h (2), 5h (2), 41mg/dl (1).

target — 1/50 runs

Points made by a single run only: 4.

Phrases used by at least half the runs: “compensation and predictability” (1), “first and safest” (1), “risk then sequentially” (1), “excessive basal raising” (1), “0.5 u/h lower” (1), “urgent safety issue” (1), “driven by low” (1), “isf to 35” (1), “observed overnight hypoglycemia” (1), “reduce basal 00” (1), “basal 00 00” (1), “stability observed overnight” (1).

Evidence cited: 0.5u/h (1), 00:00 (1), 100mg/dl (1), 35mg/dl (1), 6h (1).

primary recommendation

basal — 18/50 runs

Shared reasoning

- (17/18) Reduce overnight basal rate (00:00–05:00).

Varying reasoning

- (8/18) Reduce overnight basal rate.
- (4/18) Lowering it to 0.5 U/h may reduce hypoglycemia risk while allowing the algorithm to increase delivery if needed.
- (3/18) Consider lowering it from 0.6 to 0.5 U/h and monitor overnight glucose closely for at least 3 nights.
- (2/18) Reducing it will lower hypoglycemia risk while maintaining basal coverage.

Points made by a single run only: 23.

Phrases used by at least half the runs: “overnight basal” (18), “basal rate” (15), “reduce overnight basal” (15), “0.6 u/h” (14), “overnight basal rate” (13), “0.5 u/h” (11).

Evidence cited: 05:00 (17), 00:00 (17), 0.6u/h (14), 0.5u/h (10), 18.5% (3), 0.55u/h (2), 01:00 (2), 0.1u/h (2), 5u/h (1), 02:00 (1), 70mg/dl (1), 18% (1).

isf — 16/50 runs

Varying reasoning

- (6/16) Increase Insulin Sensitivity Factor (ISF).
- (3/16) The profile ISF of 56 mg/dL/U, combined with Dynamic ISF adjustment factor 70%, results in an effective ISF around 30-40 mg/dL/U in many periods.
- (3/16) Multiple correction bolus responses show that the current ISF of 56 mg/dL/U leads to excessive glucose drops.
- (2/16) Multiple therapy-response windows demonstrate that small correction boluses cause disproportionately large glucose drops (e.g., 0.2 U → −178 mg/dL).
- (2/16) Raising ISF to 150 mg/dL/U will reduce bolus doses and hypoglycemia risk.
- (2/16) Over multiple correction boluses, glucose drops far exceed the profile ISF prediction.
- ... 3 more varying points

Points made by a single run only: 37.

Phrases used by at least half the runs: “56 mg/dl/u” (15), “isf of 56” (12), “dynamic isf” (9), “insulin sensitivity” (8), “hypoglycemia risk” (8), “profile isf” (8).

Evidence cited: 56mg/dl (15), 40mg/dl (4), 70mg/dl (4), 70% (3), 178mg/dl (2), 100mg/dl (2), 30mg/dl (1), 0.65u (1), 150mg/dl (1), 36mg/dl (1), 0.2u (1), 110mg/dl (1).

aaps_safety_limits — 2/50 runs

Shared reasoning

- (2/2) Reduce max IOB and max basal limits.

Points made by a single run only: 3.

Phrases used by at least half the runs: “reduce max iob” (2), “limits lower max_iob_units” (1), “usage and provide” (1), “max basal limits” (1), “iob and max” (1), “max_basal_u_per_hour from 12” (1), “better match typical” (1), “12 u/h” (1), “match typical insulin” (1), “provide safety ceiling” (1), “max_iob_units from 25” (1), “basal limits lower” (1).

Evidence cited: 25u (2), 5u (1), 12u/h (1), 2u/h (1), 12u (1).

dia — 2/50 runs

Points made by a single run only: 6.

Phrases used by at least half the runs: “dia to hours” (2), “corrections causing overnight” (1), “overall insulin burden” (1), “10 hour dia” (1), “overnight hypoglycemia reducing” (1), “causing overnight hypoglycemia” (1), “lower overall insulin” (1), “hypoglycemia reducing dia” (1), “reduce dia” (1), “small corrections causing” (1), “multiple small corrections” (1), “iob to decay” (1).

Evidence cited: 6h (1), 5h (1), 10h (1).

meal_strategy — 1/50 runs

Points made by a single run only: 5.

Phrases used by at least half the runs: “subsequent hypoglycemia pre” (1), “diminish the iob” (1), “stacking that drives” (1), “causes subsequent hypoglycemia” (1), “insulin and stacked” (1), “lower total insulin” (1), “00 and evening” (1), “boluses will reduce” (1), “hyperglycemia uam responds” (1), “root cause” (1), “responds with delayed” (1), “pre meal boluses” (1).

Evidence cited: 16:00 (1), 13.2g/u (1), 10.5g/u (1), 08:00 (1), 20:00 (1).

data_foundation — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “carbohydrate intake consistently” (1), “timing and smb/uam” (1), “record all carbohydrate” (1), “timing or meal” (1), “logging every meal” (1), “future analyses” (1), “cr bolus timing” (1), “optimize cr bolus” (1), “exist making” (1), “meal strategies start” (1), “smb/uam behavior” (1), “impossible to evaluate” (1).

aaps_safety — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “daily insulin requirement” (1), “total daily insulin” (1), “iob and max” (1), “12 u/h” (1), “reduce max iob” (1), “insulin requirement 20” (1), “insulin needs max” (1), “exceeds the total” (1), “nullifies critical safety” (1), “impairing loop function” (1), “20 and nullifies” (1), “aligned with actual” (1).

Evidence cited: 20u (1), 25u (1), 4u/h (1), 12u/h (1), 8u (1).

aaps — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “algorithm errors ceiling” (1), “10 provides safety” (1), “severe insulin overdosing” (1), “exceeds any observed” (1), “reduce max iob” (1), “iob from 25” (1), “provides safety margin” (1), “5.9 and creates” (1), “iob peak 5.9” (1), “sensor or algorithm” (1), “allowing full loop” (1), “risk of severe” (1).

Evidence cited: 25u (1), 10u (1), 5.9u (1).

automation_sensitivity_limits — 1/50 runs

Points made by a single run only: 4.

Phrases used by at least half the runs: “dangerous over amplification” (1), “loop from calculating” (1), “safe floor eliminating” (1), “set dynamic_isf_adjustment_factor_percent” (1), “true to preserve” (1), “prevent the loop” (1), “ability to reactivate” (1), “later with safer” (1), “insulin sensitivity” (1), “cause of severe” (1), “reactivate later” (1), “severe hypoglycemia keep” (1).

Evidence cited: 0% (1).

uam_aggressiveness — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “smb basal minutes” (1), “unannounced meal boluses” (1), “correcting leading” (1), “minutes to limit” (1), “bolus size uam” (1), “frequently over correcting” (1), “size of unannounced” (1), “limit the size” (1), “20 to 15” (1), “driven bolus size” (1), “reduce uam driven” (1), “uam driven bolus” (1).

Evidence cited: 15min (1).

UAM SMB delivery — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “aggression temporarily lower” (1), “established this reduces” (1), “minutes to limit” (1), “uam smb aggression” (1), “triggered by unannounced” (1), “unannounced carbohydrate absorption” (1), “10 minutes” (1), “20 to 10” (1), “smb aggression temporarily” (1), “stacking while meal” (1), “automated boluses triggered” (1), “reduces the magnitude” (1).

Evidence cited: 10min (1).

automation_sensitivity — 1/50 runs

Points made by a single run only: 4.

Phrases used by at least half the runs: “isf of 56” (1), “100 will restore” (1), “preventing excessive iob” (1), “least days” (1), “70 factor” (1), “dynamic isf adjustment” (1), “boluses and preventing” (1), “severe lows increasing” (1), “size of automated” (1), “isf adjustment factor” (1), “excessive iob stacking” (1), “restore the static” (1).

Evidence cited: 70% (1), 100% (1), 56mg/dl (1).

meal — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “begin logging” (1), “meals the system” (1), “implement simple meal” (1), “logged meals” (1), “strategy prebolus 15” (1), “strategy without logged” (1), “15 minutes 100” (1), “bolus no carbs” (1), “dark implement simple” (1), “carbohydrates and declare” (1), “apply appropriate meal” (1), “meal strategy prebolus” (1).

Evidence cited: 15min (1), 100% (1).

target — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “target of 90” (1), “overnight glucose targets” (1), “00 to 08” (1), “raise overnight glucose” (1), “mg/dl from 00” (1), “90 mg/dl” (1), “contributes to overnight” (1), “floor without sacrificing” (1), “give the system” (1), “increasing these targets” (1), “targets to reduce” (1), “reduce hypoglycemia risk” (1).

Evidence cited: 90mg/dl (1), 00:00 (1), 100mg/dl (1), 23:00 (1), 08:00 (1).

aaps_uam — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “sensitivity lower min_” (1), “uam driven meal” (1), “lower min_ m_carbimpact” (1), “rapid smb deliveries” (1), “driven meal detection” (1), “triggering rapid smb” (1), “detection sensitivity lower” (1), “detect meals triggering” (1), “high threshold causes” (1), “post hyper hypoglycaemia” (1), “contribute to severe” (1), “reduce uam driven” (1).

Evidence cited: 5min (1), 5mg/dl (1).

summary

50 runs

Varying reasoning

- (17/50) Over 14 days, CGM coverage was excellent (98.7%).
- (12/50) Over the 14-day period, glucose time-in-range was 86% with significant hypoglycemia (TBR 5.1%, including severe lows of 41–42 mg/dL).
- (11/50) The main pattern is overnight hypoglycemia, especially between 01:00 and 03:00 local time (TBR up to 18.5%), and afternoon hyperglycemia (16:00-17:00 TAR 32%).
- (8/50) The current basal rate of 0.6 U/h from 00:00 to 05:00 appears slightly too high for this period.
- (8/50) No carbohydrate entries were logged, blocking meal-strategy evaluation.
- (7/50) No meal or carbohydrate entries exist, preventing evaluation of carb ratios, pre-bolus timing, or e-carb strategies.
- ... 31 more varying points

Points made by a single run only: 130.

Phrases used by at least half the runs: “carbohydrate entries” (27), “14 day” (26).

Evidence cited: 98.7% (28), 5.1% (27), 86% (27), 18.5% (18), 41mg/dl (17), 86.0% (16), 56mg/dl (16), 25u (13), 01:00 (13), 00:00 (9), 8.9% (8), 0.6u/h (8).

issues

50 runs

Varying reasoning

- (19/50) Recurrent overnight hypoglycemia (TBR up to 18.5% at 01:00) with BG as low as 41 mg/dL.
- (15/50) Absence of any carb entries prevents evaluation of carb ratios and meal bolus strategy.
- (8/50) Max basal (12 U/h) and max IOB (25 U) safety limits are far above clinical needs, posing a risk of severe overdose if misapplied.
- (8/50) High glucose variability (CV 32.6%) suggests mismatch between insulin delivery and needs.
- (7/50) Profile ISF of 56 mg/dL/U is well below observed insulin sensitivity, forcing Dynamic ISF to constantly compensate.
- (6/50) Dynamic ISF adjustment factor of 70% creates effective ISF as low as 27.8 mg/dL/U, causing over-aggressive corrections.
- ... 25 more varying points

Points made by a single run only: 82.

Phrases used by at least half the runs: “41 mg/dl” (25).

Evidence cited: 41mg/dl (25), 18.5% (23), 25u (19), 56mg/dl (18), 10h (12), 01:00 (12), 42mg/dl (11), 70% (10), 32.6% (9), 12u/h (8), 5.1% (7), 15.3% (5).

step findings: data_quality

50 runs

Shared reasoning

- (37/50) CGM coverage is 98.7% with 10 gaps (largest 65 min).

Varying reasoning

- (17/50) CGM coverage 98.7% with 3995 valid entries at ~5-minute cadence; largest gap 65 minutes.
- (17/50) Telemetry includes glucose, treatments, device statuses, loop statuses, and therapy response windows.
- (14/50) AAPS profile is complete and declared applicable for the period.
- (8/50) No carbohydrate entries exist.
- (8/50) No invalid or duplicate entries.
- (7/50) AAPS profile and settings are complete.
- ... 16 more varying points

Points made by a single run only: 24.

Phrases used by at least half the runs: “cgm coverage” (45), “cgm coverage 98.7” (35), “aaps profile” (28), “65 min” (27).

Evidence cited: 98.7% (47), 65min (42), 5min (12), 275.95u (1), 24h (1).

step findings: safety_overview

50 runs

Varying reasoning

- (25/50) Overnight hours (00:00–02:00 local) show high TBR (up to 18.5%).
- (13/50) Overall TIR 86.0%, TAR 8.9%, TBR 5.1%.
- (10/50) Severe lows 41-42 mg/dL on Aug-13, Aug-14, Aug-21.
- (10/50) TIR 86%, TAR 8.9%, TBR 5.1%, CV 32.6%.
- (9/50) Severe lows recorded: 41, 42, 42 mg/dL.
- (9/50) Afternoon TAR up to 32.1% at 16:00 local.
- ... 29 more varying points

Points made by a single run only: 55.

Phrases used by at least half the runs: “cv 32.6” (27), “41 mg/dl” (26), “tbr 5.1” (26).

Evidence cited: 5.1% (40), 32.6% (36), 18.5% (34), 41mg/dl (26), 42mg/dl (24), 8.9% (23), 86.0% (14), 01:00 (14), 32.1% (13), 15.3% (12), 00:00 (12), 293mg/dl (11).

step findings: basal_targets

50 runs

Shared reasoning

- (32/50) Overnight hypoglycemia (TBR 18.5% at 01:00-02:00) despite frequent zero-temp basals indicates basal rate (0.6 U/h 00:00-05:00) may be too high.

Varying reasoning

- (13/50) Target of 90 mg/dL overnight is appropriate.
- (7/50) Targets of 90-100 mg/dL are reasonable.
- (6/50) Evening basal 0.6–0.7 U/h contributes to a TBR of 15.3% at 19:00.

- (5/50) Afternoon basal 16:00-18:00 (0.5 U/h) may contribute to afternoon hyperglycemia, but meal effects cannot be excluded.
- (4/50) Afternoon basal (0.5 U/h 16:00–18:00) appears insufficient, with average >160 mg/dL.
- (3/50) Early morning (05:00-09:00): glucose rises from ~99 to 133 mg/dL, well controlled by basal steps at 0.7 and 0.8 U/h.
- ... 12 more varying points

Points made by a single run only: 109.

Phrases used by at least half the runs: “0.6 u/h” (31), “90 mg/dl” (25).

Evidence cited: 0.6u/h (31), 00:00 (29), 90mg/dl (25), 05:00 (23), 0.5u/h (19), 18.5% (19), 0.8u/h (17), 0.7u/h (17), 16:00 (16), 100mg/dl (12), 01:00 (11), 18:00 (11).

step findings: profile_cr_isf_dia

50 runs

Varying reasoning

- (19/50) Carb ratios cannot be evaluated.
- (17/50) CR cannot be evaluated without carb entries.
- (7/50) Correction-only boluses show insulin effect far exceeding the 56 mg/dL/U ISF (e.g., 0.2 U → -178 mg/dL).
- (7/50) DIA of 10 hours appears appropriate.
- (6/50) DIA of 10 hours is longer than typical for rapid-acting insulin and may contribute to IOB stacking and delayed hypoglycemia.
- (4/50) Profile ISF (56 mg/dL/U) appears too low given correction overshoots, but dynamic ISF amplifies the effective potency.
- ... 18 more varying points

Points made by a single run only: 82.

Phrases used by at least half the runs: “56 mg/dl/u” (45), “dynamic isf” (31), “cannot be evaluated” (25).

Evidence cited: 56mg/dl (45), 10h (45), 0.2u (11), 70% (6), 178mg/dl (5), 13.2g/u (3), 100mg/dl (3), 125mg/dl (3), 40mg/dl (2), 53mg/dl (2), 0.15u (2), 70mg/dl (2).

step findings: automation_smb_uam

50 runs

Varying reasoning

- (15/50) SMB strength (interval 3 min, max 15 min basal) and UAM max 20 min basal are conservative.
- (14/50) SMB and UAM are enabled;
- (7/50) SMB activation conditions are consistent: SMB always enabled, SMB with COB and after-carbs disabled, SMB with temp target enabled.
- (5/50) SMB always enabled, UAM enabled.
- (4/50) Activation conditions are appropriate.
- (3/50) Max basal (12 U/h) and max IOB (25 U) limits are not binding.
- ... 7 more varying points

Points made by a single run only: 118.

Phrases used by at least half the runs: “smb and uam” (28).

Evidence cited: 20min (14), 3min (12), 25u (7), 15min (5), 5.9u (3), 12u/h (3), 5min (3), 8mg/dl (3), 20u (2), 0.27u (2), 0.3u (2), 66mg/dl (1).

step findings: automation_sensitivity_limits

50 runs

Shared reasoning

- (35/50) Safety limits (max basal 12 U/h, max IOB 25 U) are not restrictive.

Varying reasoning

- (15/50) Dynamic ISF is enabled with an adjustment factor of 70%.
- (14/50) Autosens is disabled.
- (8/50) Dynamic ISF is heavily compensating for a mismatched ISF profile, with adjustment factor 70% leading to effective ISF as low as 27.8 mg/dL/U.
- (3/50) The max IOB limit of 25 U exceeds total daily insulin and poses a latent safety risk; no events have approached this limit.
- (3/50) Carbohydrate absorption model (min_5m_carbimpact 8 mg/dL/5min, meal_max_absorption_hours 7 h) cannot be tested without carb data.
- (2/50) The dynamic algorithm correctly detects high sensitivity (variable_sens often <50) but its influence is dampened, so blended ISF remains too low.
- ... 12 more varying points

Points made by a single run only: 92.

Phrases used by at least half the runs: “dynamic isf” (48), “max iob” (41), “12 u/h” (37), “max basal” (37), “iob 25” (33), “max iob 25” (32), “basal 12 u/h” (30), “max basal 12” (30).

Evidence cited: 25u (42), 70% (39), 12u/h (37), 27.8mg/dl (8), 5.9u (5), 56mg/dl (4), 20u (4), 7h (3), 8mg/dl (3), 3.6u/h (3), 6u (3), 124mg/dl (2).

step findings: meal_bolus_strategy

50 runs

Varying reasoning

- (25/50) Meal bolus timing, immediate-bolus percentage, and CR cannot be evaluated.
- (15/50) Zero carbohydrate entries exist.
- (10/50) The system relies entirely on UAM detection.
- (8/50) No declared meal strategy and zero carb entries.
- (5/50) Meal strategies are not declared (analysis_context.meal_strategies empty).
- (4/50) No carbohydrate entries exist in the 14-day period.
- ... 12 more varying points

Points made by a single run only: 62.

Phrases used by at least half the runs: “immediate bolus” (29), “bolus percentage” (25).

Evidence cited: 100% (3), 13.2g/u (2), 15min (2), 09:00 (1), 14:00 (1), 07:00 (1), 22:00 (1), 30min (1), 10.5g/u (1), 12:00 (1), 26.3% (1), 20:00 (1).

step findings: meal_ecarbs_absorption

50 runs

Varying reasoning

- (18/50) No extended carb entries exist.
- (14/50) Absorption model settings (min_5m_carbimpact 8 mg/dL/5 min, meal_max_absorption_hours 7 h) cannot be validated without meal and carb data.
- (9/50) The carbohydrate absorption parameters cannot be evaluated.
- (7/50) No e-carb or regular carb entries exist, so absorption patterns cannot be assessed.
- (4/50) No e-carb declarations or treatments.
- (3/50) E-carb recommendations are impossible without meal data.
- ... 9 more varying points

Points made by a single run only: 40.

Phrases used by at least half the runs: “carb entries” (26), “absorption model” (25).

Evidence cited: 7h (12), 5min (7), 8mg/dl (7).

step findings: synthesis_plan

50 runs

Varying reasoning

- (15/50) 3) Reduce basal 00:00-05:00 from 0.6 to 0.5 U/h;
- (10/50) Automation settings should remain unchanged.
- (6/50) Carbohydrate logging is essential for future meal strategy adjustments.
- (5/50) Safety limits (max IOB 25 U, max basal 12 U/h) are hazardous.
- (5/50) Introducing carbohydrate logging for one meal per day will enable future CR and bolus optimization.
- (5/50) Meal strategies cannot be addressed until carbohydrate logging begins.
- ... 24 more varying points

Points made by a single run only: 155.

Evidence cited: 00:00 (17), 05:00 (15), 0.6u/h (14), 56mg/dl (13), 25u (8), 0.5u/h (6), 70% (6), 10h (5), 90mg/dl (4), 70mg/dl (4), 16:00 (3), 0.55u/h (3).

meal strategy

50 runs

Varying reasoning

- (13/50) Begin logging all carbohydrate intake immediately.
- (11/50) Until meal data are available, no prebolus timing, immediate bolus percentage, or e-carb strategy can be recommended.
- (7/50) Zero carbohydrate entries prevent any meal-pattern analysis.
- (6/50) Start recording at least one meal per day with accurate carbohydrate amounts.
- (6/50) Start with a short pre-bolus of 10-15 minutes, 100 % immediate bolus, and keep SMB/UAM enabled only after the ISF has been corrected to 36 mg/dL/U.
- (6/50) Zero carbohydrate entries exist in the 14-day period, and the user has not declared a manual bolus strategy.

- ... 31 more varying points

Points made by a single run only: 175.

Phrases used by at least half the runs: "immediate bolus" (34).

Evidence cited: 100% (18), 15min (10), 20min (4), 10min (2), 5u (2), 16:00 (1), 13.2g/u (1), 12.6g/u (1), 18% (1), 00:00 (1), 10.5g/u (1), 15% (1).